DATA ENGINEERING / PREDICTIVE MODELING

FORMULA 1

Podium Prediction

A reproducible, leakage-conscious study of driver-race data, historical race trends, three predictive models, controlled preprocessing experiments, ablation and explainable AI.

ACL | Milestone 1

ANALYSIS PERIOD	TARGET	INFERENCE CUTOFF
1950–2024	Top-three race finish (binary)	After qualifying, before race start

Primary comparison Validation precision-recall AUC (PR-AUC)

Selected model in the original run Two-hidden-layer feedforward neural network

Dataset Formula 1 World Championship (Kaggle / Ergast), 14 relational CSVs

REPORT BASIS

Prepared from the uploaded Kaggle notebook *acl-ms1-f1-podium(2).ipynb*. Numerical narrative and recreated result charts use its documented first successful execution. An explicit reproducibility note discusses the later Run All outputs.

13 October 2026

Executive summary

This project predicts whether a Formula 1 driver will finish in the top three using only information available once qualifying is complete and before the race begins. The input consists of a driver-race analytical table engineered from 14 linked CSV files. After handling literal missing-value sentinels and historical shared-drive duplicates, the final table contains **26,668 unique driver-race observations**, including 3,395 positive podium outcomes (12.73%).

Principal result - original successful execution

The two-hidden-layer neural network achieved **validation PR-AUC 0.7393** and **unseen test PR-AUC 0.7131**. Logistic Regression was a competitive baseline and achieved the best test F1 score (0.6869). Model selection and thresholds were based on training/validation only.

Research pipeline

STAGE	EVIDENCE / OUTPUT
Audit & cleaning	14 tables checked; 91 duplicate driver-race rows resolved; PK/FK tests passed.
Historical analysis	Front-row circuit conversion, home-vs-away grid-adjusted podium rates, mechanical retirements by constructor.
Prediction	Five leakage-safe inputs; 24,109 train / 1,200 validation / 1,359 test rows.
Experiments	Four preprocessing orders; three model families; four feature-group ablations.
Interpretation	Permutation importance, global SHAP, local LIME, complete/missing-record inference.

Headline domain findings

- Actual starting-grid position is the strongest single predictor across exploratory analysis, feature ablation and explanatory methods.
- Grid-adjusted home and away podium rates are nearly equal (13.21% versus 13.60%).
- Historical front-row podium conversion varies substantially by circuit and pre-2014 versus 2014–2024 era; very small samples can produce 100% rates.
- The latest rerun slightly changed the ranking of validation PR-AUC between the neural network and Random Forest; this report separates the first-run narrative from the saved rerun outputs.

1. Problem definition and data

A single record represents one driver entering one race. The binary target is **podium = 1** when the final position order is at most three, and **podium = 0** otherwise. Data from the race itself - finishing position, pit stops, completed laps, finishing points and race status - are not available at the inference cutoff and were excluded as features.

The source is the Kaggle *Formula 1 World Championship (1950–2020)* dataset, whose exported tables in this study cover seasons through 2024. The relational collection has 14 tables: circuits, constructors, constructor_results, constructor_standings, drivers, driver_standings, lap_times, pit_stops, qualifying, races, results, seasons, sprint_results and status.

2. Data auditing, cleaning and EDA

2.1 Initial inspection and missingness

All 14 tables were loaded as strings to retain literal missing-value placeholders for auditing. The study computed table sizes, sentinel counts, exact duplicate rows, and repeated pairs of raceId and driverId in the results table. Missingness is substantial in selected historical columns because Formula 1 record-keeping is not uniform across decades.

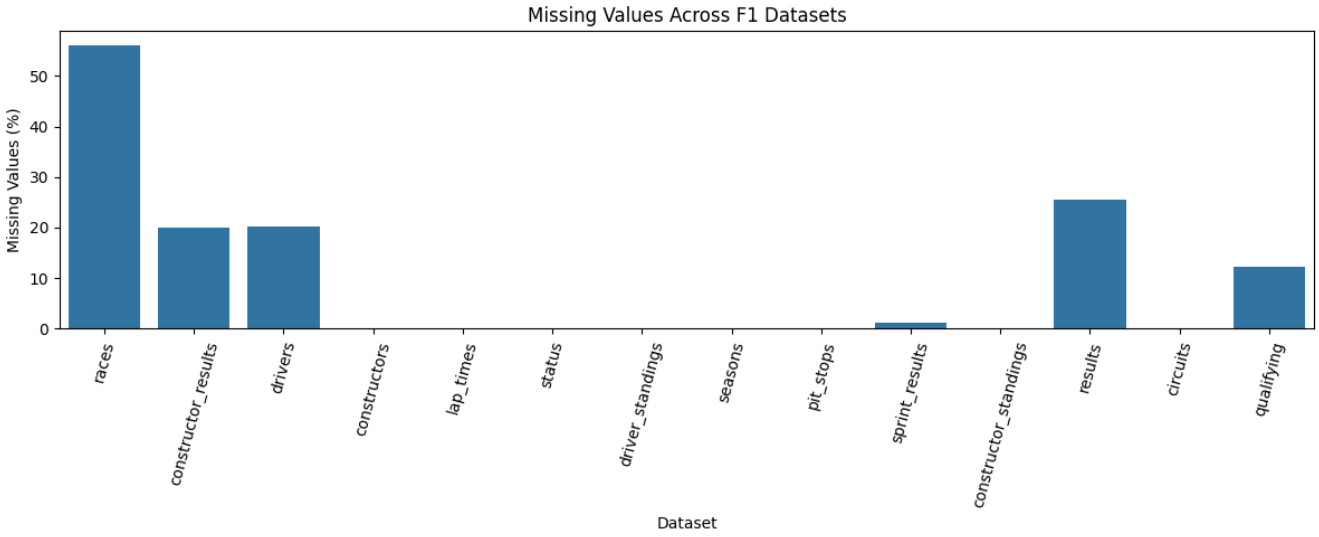


Figure 1. Percentage of missing placeholders by original dataset table (notebook initial EDA).

2.2 Cleaning rules and relational integrity

Raw CSVs were preserved. Working copies replaced **\N** and empty strings with NaN, coerced numeric fields, converted date fields, and derived seconds from lap/qualifying/pit-stop duration strings. Primary-key validation tested 14 table key definitions for nulls and repeated keys; all recorded zero violations. The notebook also tested 23 foreign-key references against parent tables and found no invalid non-null references.

Historical shared drives create multiple result rows for the same driver in the same race. The pipeline computed the podium label as the maximum top-three indicator within each repeated pair, retained the row with the lowest positive grid position (using resultId to break ties), and enforced uniqueness on (raceId, driverId). This reduced 26,759 original results rows to **26,668**, removing 91 duplicates.

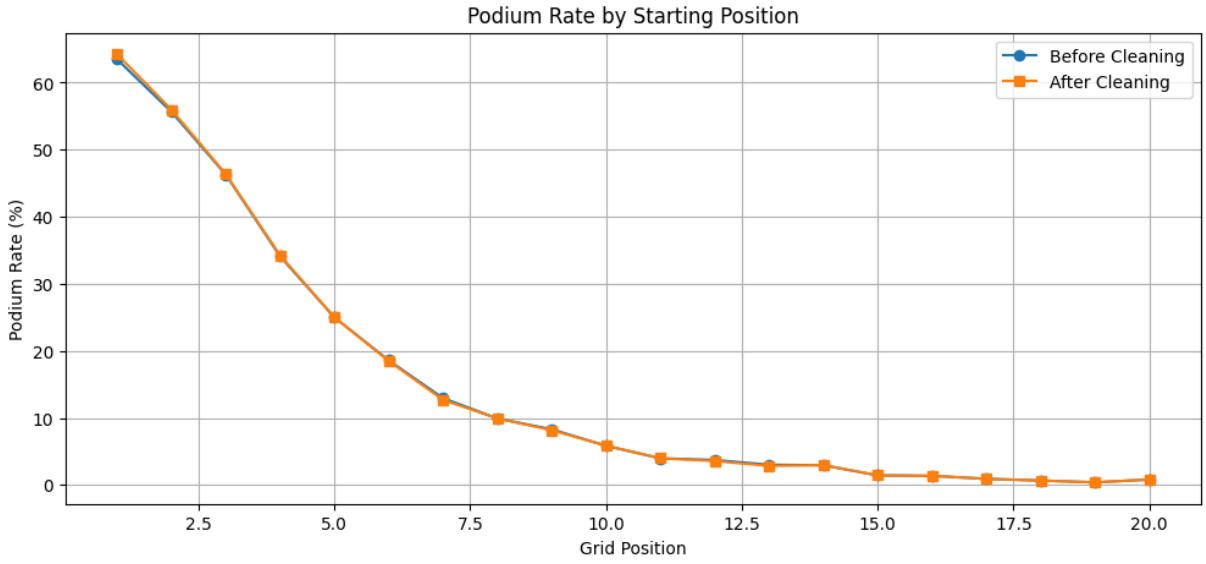


Figure 2. The starting-grid / podium relationship is nearly unchanged by duplicate resolution.

Data-audit outcome

The resulting modeling grain is exactly one row per (raceId, driverId). Zero duplicate pairs remain. Missing-value replacement changes the representation of missing data, not the count of missing observations.

2.3 Outcome distribution and exploratory findings

There are 23,273 non-podium observations and 3,395 podium observations. The **12.73%** positive prevalence makes accuracy alone insufficient for evaluation. The initial plots show that early grid positions have substantially higher podium conversion than lower starting positions.

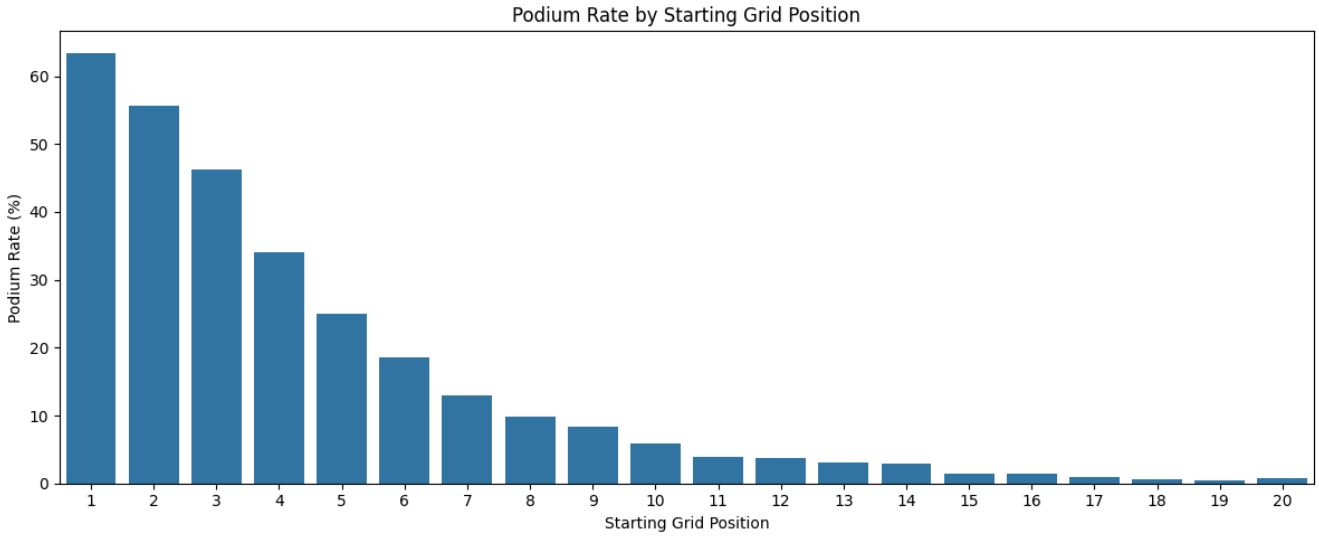


Figure 3. Podium probability by starting-grid position in the original exploratory analysis.

3. Data-engineering analysis

3.1 Q1 - Which circuits convert front-row starts into podiums?

The notebook joins **results → races → circuits** and left-joins qualifying on (raceId, driverId). A front-row starter is explicitly defined by **actual grid position 1 or 2**, not qualifying position, because penalties can change the starting order. Conversion is the share of such starters finishing in the top three. Circuits are compared before 2014 versus 2014–2024.

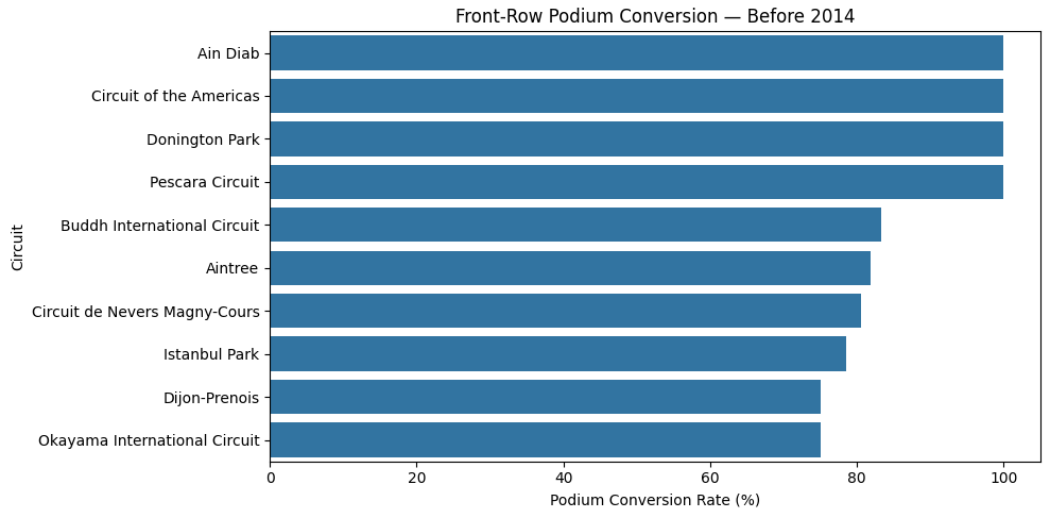


Figure 4. Top ten front-row podium conversion rates before 2014. Several 100% rates come from only 2–4 starts.

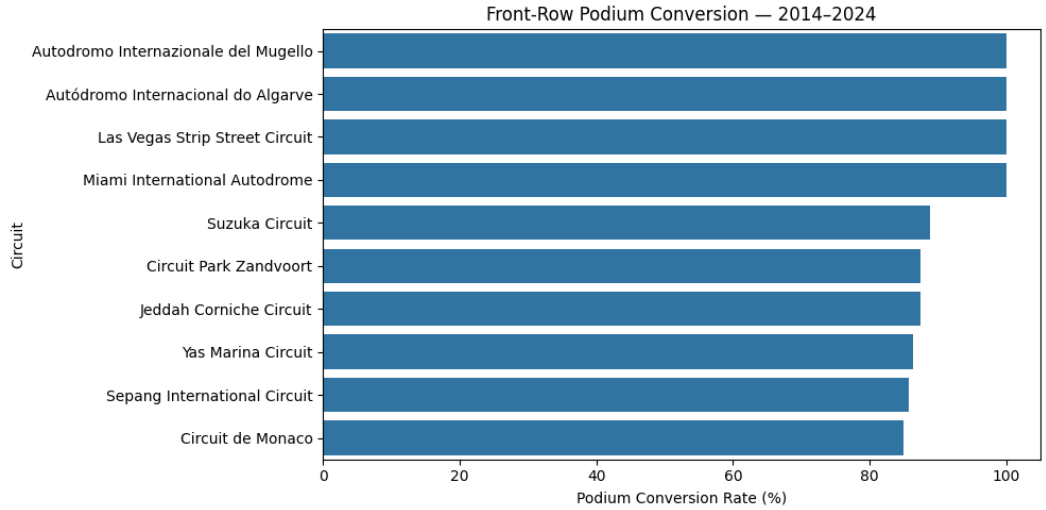


Figure 5. Top ten front-row podium conversion rates in 2014–2024.

Circuits observed in both eras

CIRCUIT	BEFORE 2014	2014–2024	CHANGE
Monaco	49.59%	85.00%	+35.41 pp
Spa-Francorchamps	54.35%	81.82%	+27.47 pp
Zandvoort	58.33%	87.50%	+29.17 pp

A total of **2,422 (23.27%)** records with both valid grid and qualifying positions had different positions. Thus, the distinction between the post-qualifying position and the race starting grid materially matters. Rankings changed across eras, but the descriptive differences should not be interpreted as causal evidence of the 2014 technical regulations; samples, drivers, teams and race environments also changed.

3.2 Q2 - Is there a home-country podium advantage?

The notebook links the driver nationality to the circuit country through **results → drivers** and **results → races → circuits**. Country names were normalized to compare driver nationality with venue country. Unmatched nationalities or unavailable/nonpositive grid positions were excluded.

A stratified comparison controls descriptively for starting position using grid bands 1–3, 4–6, 7–10, 11–15 and 16+. Both home and away podium rates were standardized to the same pooled band distribution; these are descriptive adjustments, not a causal estimate.

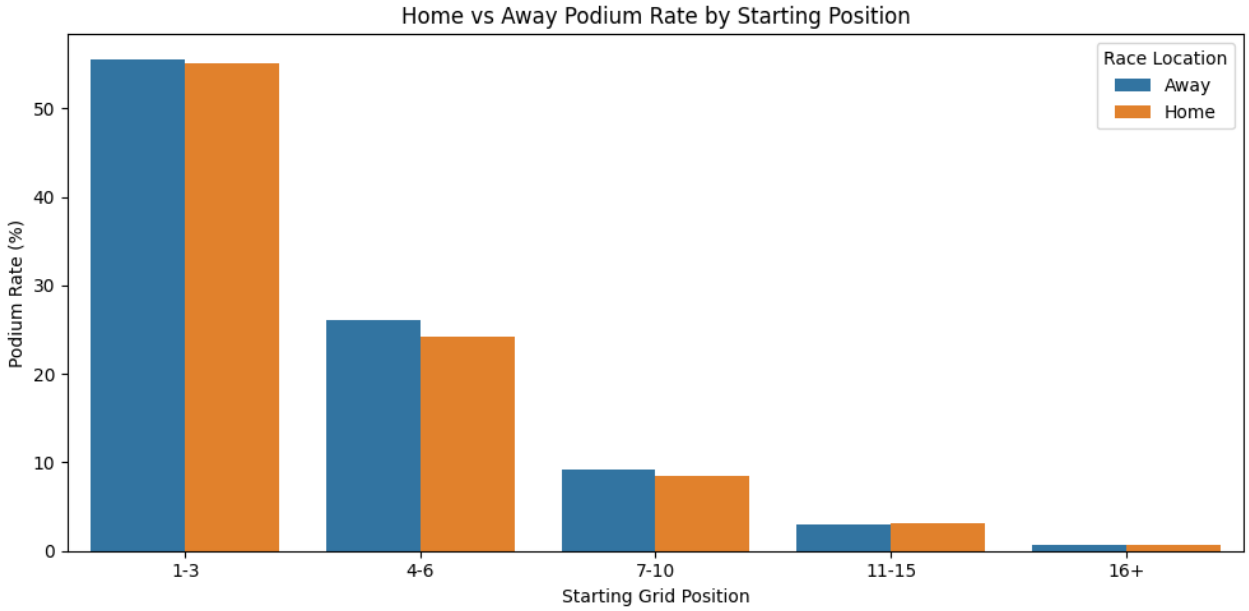


Figure 6. Home-versus-away podium rates within the same starting-grid position bands.

ADJUSTED RATE	HOME	AWAY	DIFFERENCE
Grid-standardized podium rate	13.21%	13.60%	-0.39 pp

The adjusted rate difference is small and slightly favors away races. The evidence does **not** support a clear home-country advantage after accounting for starting grid band. Nationality mappings and unequal coverage remain limitations.

3.3 Q3 - Which constructors have the most mechanical retirements?

The analysis joins **results → races, constructors and status** for 2014–2024. Status descriptions matching a documented list of component failures (engine, gearbox, hydraulics, power loss and related terms) are classified as mechanical; explicit nonstarters, nonqualifiers and withdrawals are removed. Accidents, collisions and penalties are not in the failure keyword list. Mechanical retirement rate is the number of classified mechanical retirements divided by race starters for each constructor-period group.

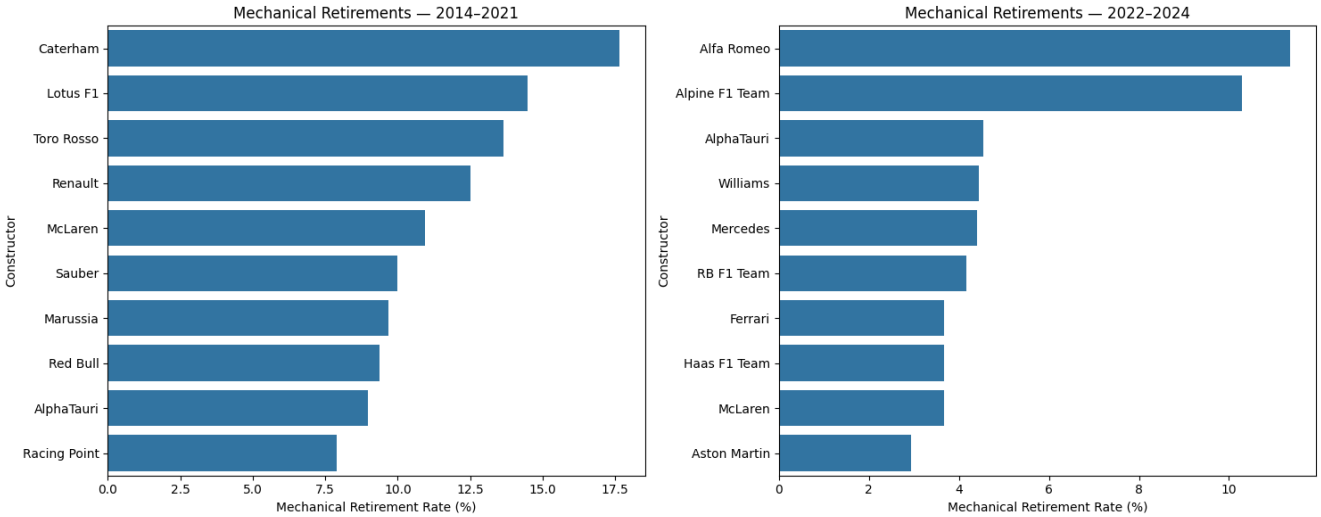


Figure 7. Highest mechanical-retirement rates by constructor across the two required time periods.

PERIOD	CONSTRUCTOR	MECHANICAL / STARTS	RATE
2014–2021	Caterham	6 / 34	17.65%
2014–2021	Lotus F1	11 / 76	14.47%
2014–2021	Toro Rosso	33 / 242	13.64%
2022–2024	Alfa Romeo	10 / 88	11.36%
2022–2024	Alpine F1 Team	14 / 136	10.29%
2022–2024	AlphaTauri	4 / 88	4.55%

Caterham had the highest measured rate in 2014–2021; Alfa Romeo led in 2022–2024. These descriptive rates are sensitive to denominators, constructor naming across eras, and the fact that free-text retirement reasons cannot always unambiguously distinguish technical failure from other causes.

4. Feature engineering and leakage controls

Exactly five original features were selected, based on the exploratory relationship with podium outcomes and their availability at the pre-race cutoff. The target was kept separate from predictors.

FEATURE	CONSTRUCTION	WHY PRE-RACE SAFE
Starting grid	Actual starting slot for the current race.	Available once the race grid is confirmed.
Qualifying position	Qualifying classification before the race.	Available after qualifying; differs from grid if penalties apply.
Historical driver form	Mean podium indicator across up to five previous races.	Driver-wise shift(1) prevents using current-race result.
Previous driver standing	Driver position in championship after prior race of same season.	Joined via previous_raceId, not current post-race standings.
Previous constructor standing	Team championship position after prior race of same season.	Joined via previous_raceId, not current post-race standings.

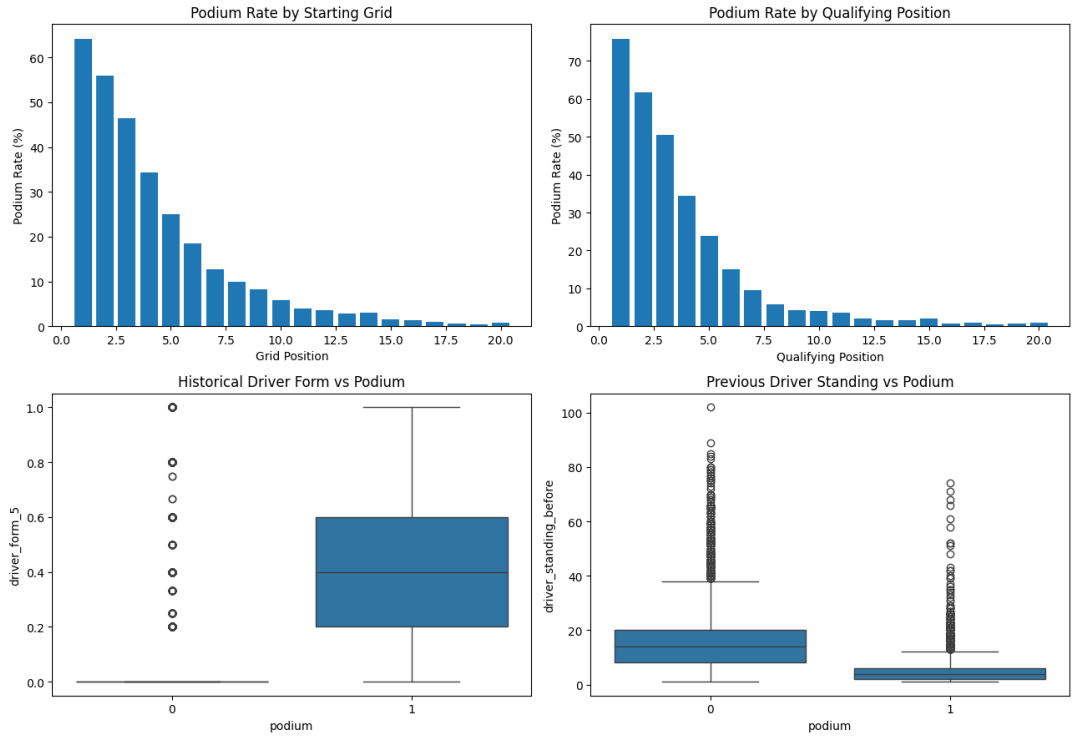


Figure 8. Empirical rationale: podium rates by grid and qualifying positions, previous driver form and previous standings.

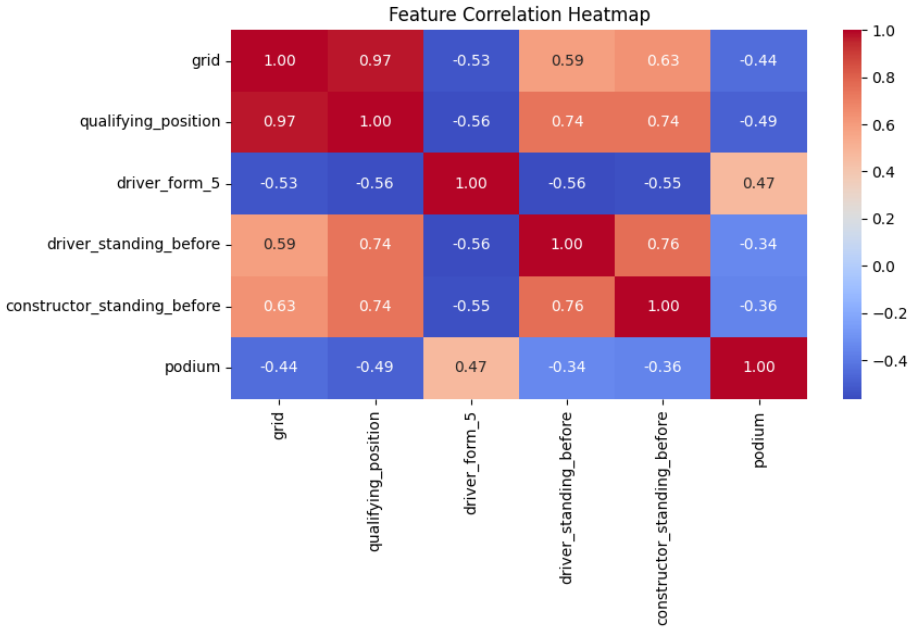


Figure 9. Correlations among selected features and target; grid and qualifying are strongly redundant (r ≈ +0.97).

4.1 Missing-value profile

MODEL FEATURE	MISSING VALUES	SHARE OF 26,668
grid	1,635	6.13%
qualifying_position	16,174	60.65%
driver_form_5	861	3.23%
driver_standing_before	3,661	13.73%
constructor_standing_before	3,524	13.21%

Historical qualifying coverage is the most important missingness issue. First-time drivers lack a five-race history; the first race of a season lacks a previous-round championship position. These gaps were deliberately retained and handled downstream rather than deleting large parts of the historical dataset.

5. Splitting and preprocessing experiments

5.1 Chronological evaluation protocol

The data was separated by season rather than randomly partitioned. Training covers **1950–2018** (24,109 records, 12.49% podium), validation **2019–2021** (1,200, 15.00%), and held-out test **2022–2024** (1,359, 15.01%). Chronological splitting is appropriate because real deployment predicts future events from earlier historical information. Preprocessing was fitted on training records only.

SPLIT	YEARS	RECORDS	PODIUM RATE
Training	1950–2018	24,109	12.49%
Validation	2019–2021	1,200	15.00%
Unseen test	2022–2024	1,359	15.01%

5.2 Controlled preprocessing-order comparisons

Four pipelines were tested using the same Logistic Regression model and validation PR-AUC. Each fitted its transformations on training data only. Two orders used median imputation and standard scaling, and two added missingness indicators. Imputation with indicators provided a small, consistent gain over no-indicator approaches.

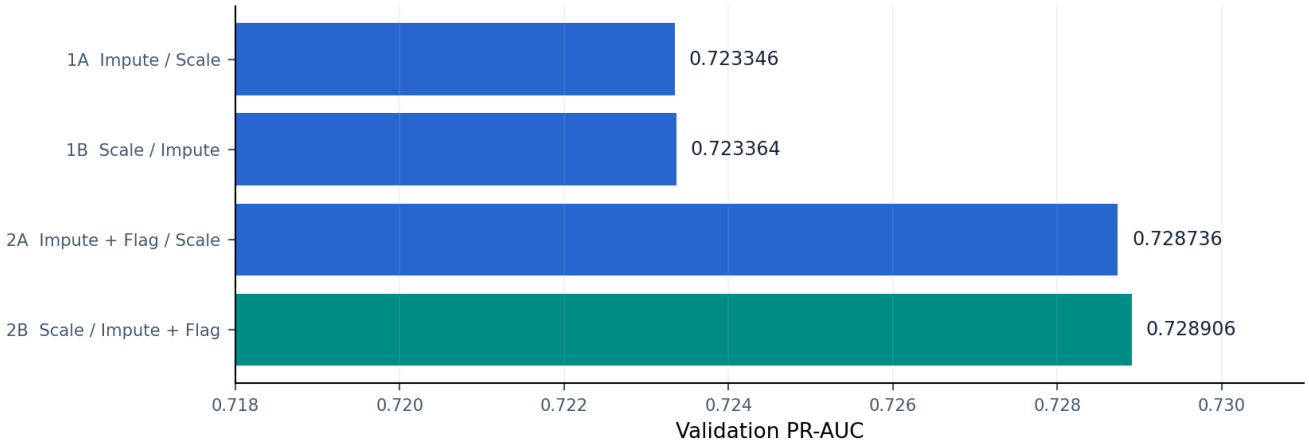


Figure 10. Validation PR-AUC of the four preprocessing pipelines; original and rerun outcomes match.

PIPELINE	ORDER	VAL. PR-AUC
1A	Impute → Scale	0.723346
1B	Scale → Impute	0.723364
2A	Impute + indicator → Scale	0.728736
2B (selected)	Scale → Impute + indicator	0.728906

Pipeline **2B** was selected by maximum validation PR-AUC, though the 2A-versus-2B difference (0.000170) is too small to claim a robust advantage of order alone. The chosen pipeline produces **10 effective inputs**: five standardized original variables and five missing-value indicators. It yields zero missing values in the preprocessed training matrix.

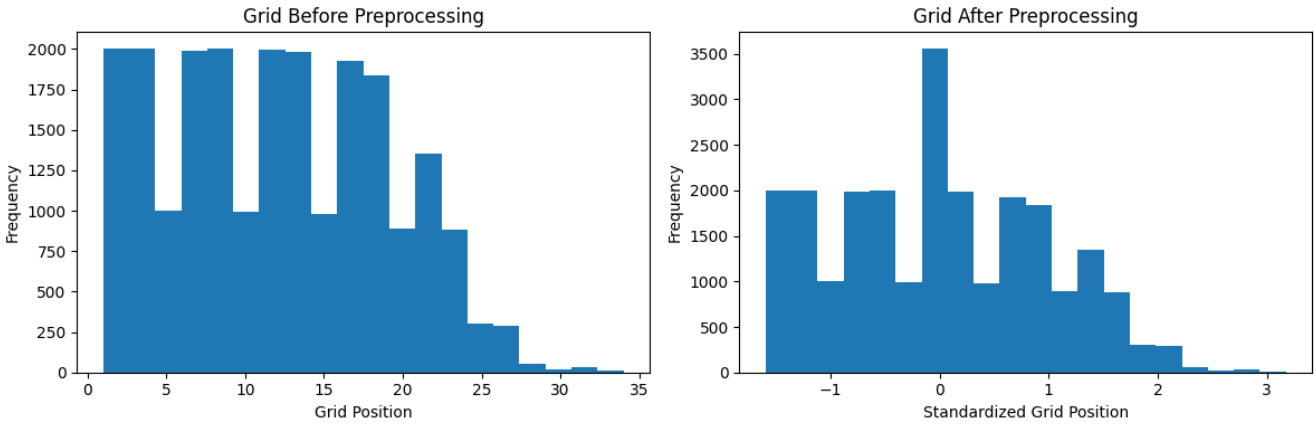


Figure 11. Grid distribution before versus after standardization and median imputation.

6. Predictive models and model selection

The benchmark compares two statistical machine-learning approaches and one shallow neural network on identical preprocessed feature inputs. The classification is binary; predicting an exact finishing position is out of scope.

MODEL	CONFIGURATION	MOTIVATION / LIMITATION
Logistic Regression (baseline)	max_iter = 2000	Simple, auditable reference; linear decision function is restrictive.
Random Forest	100 trees; max depth 10; minimum 5 cases per leaf	Captures nonlinear relations; can be harder to interpret and sensitive to era shifts.
Shallow neural network	Input 10 → Dense 32/ReLU → Dense 16/ReLU → Sigmoid 1; Adam	Learns interactions in a compact architecture; stochastic optimization and lower interpretability.

The network uses binary cross-entropy, a batch size of 32, up to 100 epochs and early stopping based on validation loss with patience 10 and restoration of the best weights. Its configuration is a justified initial design, not the result of exhaustive hyperparameter optimization.

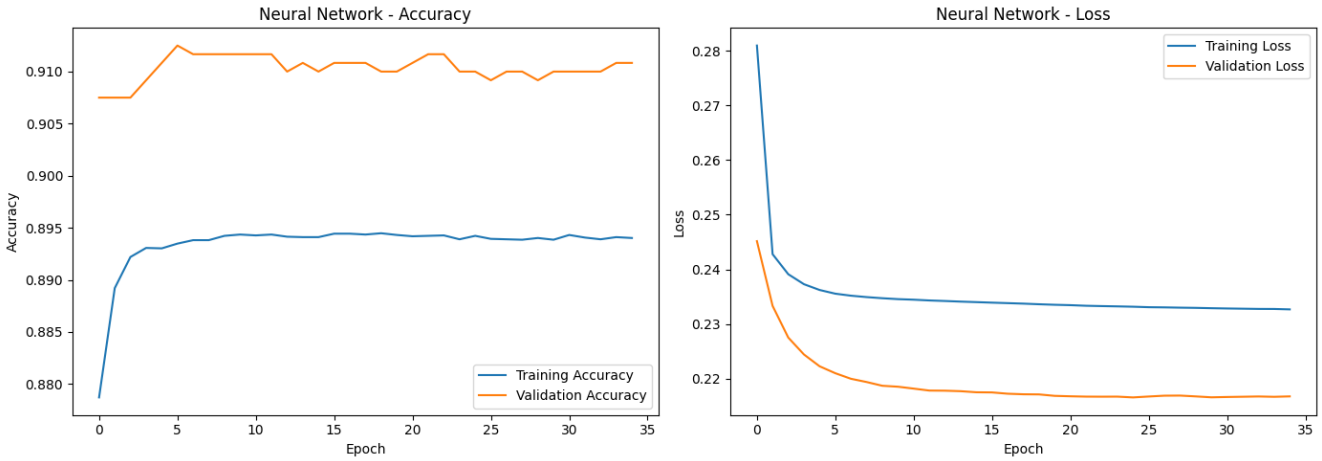


Figure 12. Network accuracy and loss during a later Run All execution (illustrative; stochastic curve values may differ).

6.1 Metrics and threshold policy

PR-AUC is the primary selection metric because there are far fewer podium than non-podium cases. **ROC-AUC** is the secondary measure of ranking discrimination. F1, precision and recall characterize the threshold-dependent decision trade-off. For each model, the classification threshold was chosen by maximizing F1 on the validation partition alone. The test dataset was never used for threshold selection.

6.2 Original-execution results

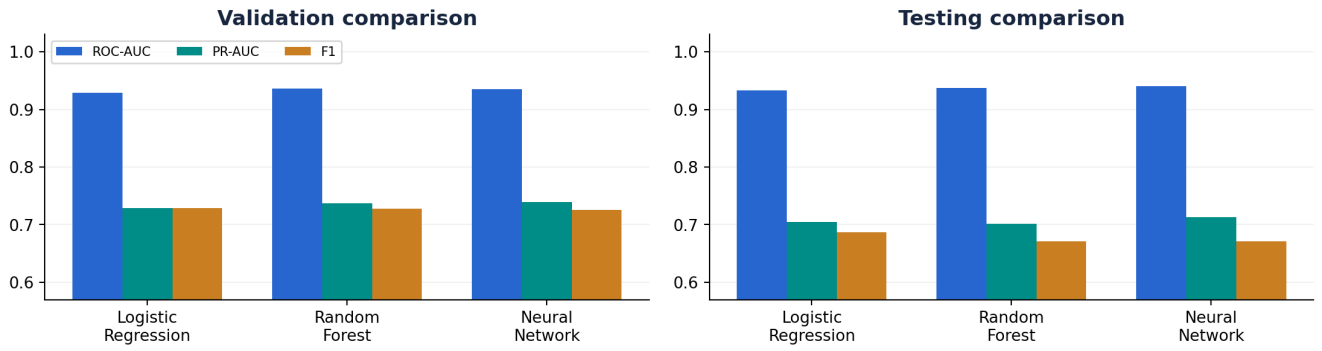


Figure 13. Model comparison recreated from the numeric results documented for the first successful execution.

MODEL	DATA	ROC-AUC	PR-AUC	F1	PREC.	RECALL
Logistic Reg.	Training	0.9036	0.5624	0.5803	0.5013	0.6888
Logistic Reg.	Validation	0.9292	0.7289	0.7290	0.6414	0.8444
Logistic Reg.	Testing	0.9334	0.7046	0.6869	0.5842	0.8333
Random Forest	Training	0.9230	0.6291	0.6034	0.5906	0.6167
Random Forest	Validation	0.9360	0.7372	0.7282	0.6935	0.7667
Random Forest	Testing	0.9374	0.7011	0.6713	0.6400	0.7059
Neural Network	Training	0.9088	0.5754	0.5675	0.5496	0.5865
Neural Network	Validation	0.9350	0.7393	0.7258	0.6847	0.7722
Neural Network	Testing	0.9399	0.7131	0.6715	0.6573	0.6863

Training PR-AUC was lower than validation PR-AUC across the three models, consistent with changes in historical data coverage and class prevalence across eras. Original-run validation PR-AUCs were **0.7289** (Logistic Regression), **0.7372** (Random Forest) and **0.7393** (neural network). Thus, the notebook originally selected the neural network under the predefined primary metric. Its original test PR-AUC was **0.7131**, whereas Logistic Regression delivered the higher test F1 (0.6869) and recall (0.8333).

How to interpret this comparison

The neural network was preferred for ranking rare podium outcomes, not for every downstream operating point. Logistic Regression remained competitive in threshold-dependent decisions. The differences between the nonlinear models are small and were not tested for statistical significance.

7. Feature-group ablation

Ablation tests whether a feature group contributes to predictive quality by removing it, refitting the same training-only preprocessing pipeline and retraining the same two-layer network. Four groups cover all five original features: Grid, Qualifying, Historical Form and the two championship Standings. Validation PR-AUC is compared with the full-model reference.

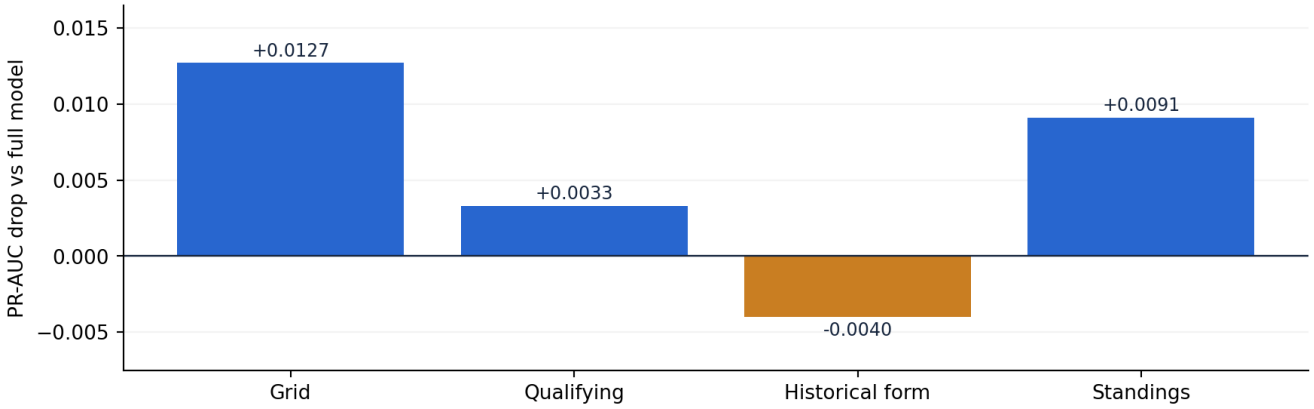


Figure 14. Ablation PR-AUC differences recreated from first-run notebook interpretations (positive = removal hurts).

REMOVED GROUP	VAL. PR-AUC	DROP FROM 0.7393
Starting grid	0.7267	+0.0127
Qualifying	0.7360	+0.0033
Historical driver form	0.7433	-0.0040
Driver + constructor standings	0.7302	+0.0091

The grid group generated the largest observed reduction in first-run validation PR-AUC, followed by previous championship standings. Qualifying adds only limited incremental information because it is highly correlated with starting grid. Removing historical form slightly improved the first-run result; repeated stochastic trials would be required to distinguish a real incremental effect from training variability.

8. Explainable AI

All explanations are applied to the fitted shallow neural network, which was the preferred model in the original run. These analyses describe how the model uses information for predictions; they do **not** identify causal drivers of on-track results.

8.1 Global permutation importance

The notebook independently permutes each original validation feature three times, reruns the fitted preprocessor and model, and records the average decline in PR-AUC. This measures how much the already trained model relies on an input under the validation distribution.

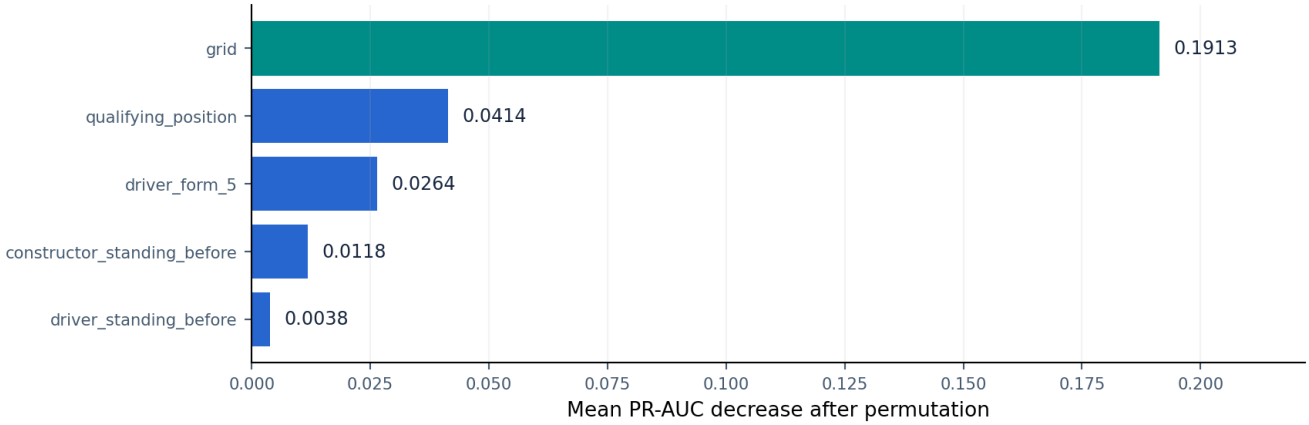


Figure 15. Permutation importance reconstructed from original-execution recorded scores.

Starting grid is dominant (mean PR-AUC loss 0.1913), followed by qualifying position (0.0414), historical driver form (0.0264), constructor standing (0.0118) and driver standing (0.0038). Correlated features can make individual permutation scores smaller than their combined predictive value.

8.2 Global SHAP

A Kernel SHAP explainer used 20 background observations sampled from training and explained 100 randomly selected validation rows with 100 evaluation samples per observation. Its 10 preprocessed inputs were collapsed into five feature groups by adding the SHAP attribution of each missingness flag to its corresponding original feature.

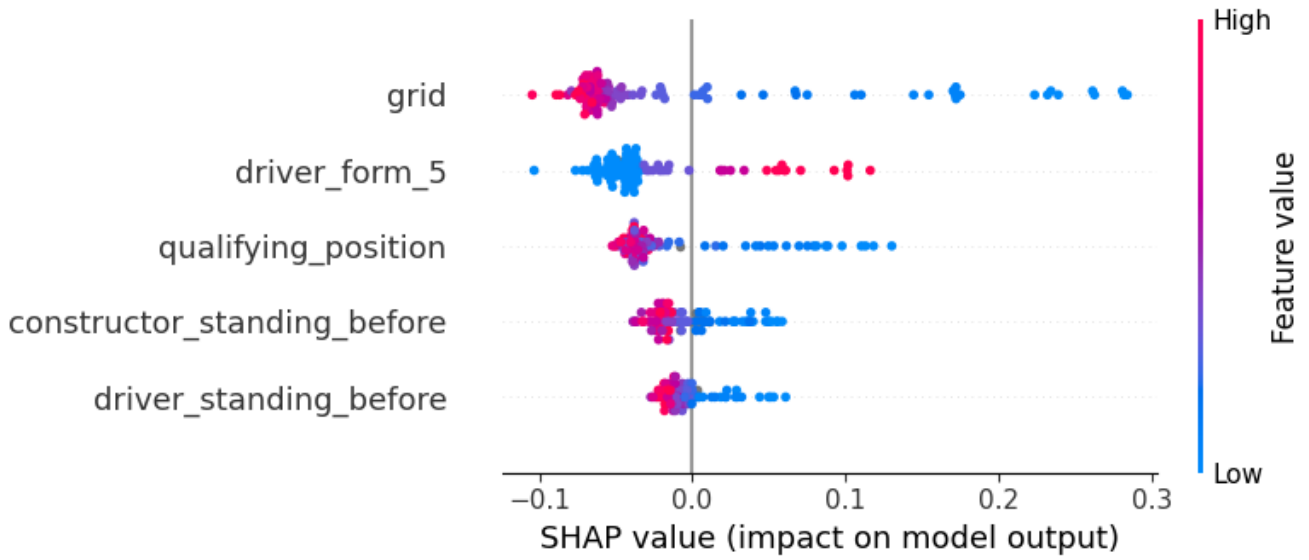


Figure 16. SHAP global summary displayed in the saved notebook rerun. Grid and recent driver form dominate the visible feature ordering.

The original interpretation ranks grid first, driver form second and qualifying third. More favorable grid slots and stronger recent form generally move predicted podium probability upward. SHAP identifies model associations conditional on the supplied background, sampling configuration and model state; the ordering need not match permutation importance or retraining-based ablation exactly.

8.3 Individual prediction explained by LIME

LIME fits a local surrogate around one driver-race validation record. Although the network accepts 10 preprocessed inputs, the explanation groups each original feature with its missing-value indicator to present five interpretable categories.

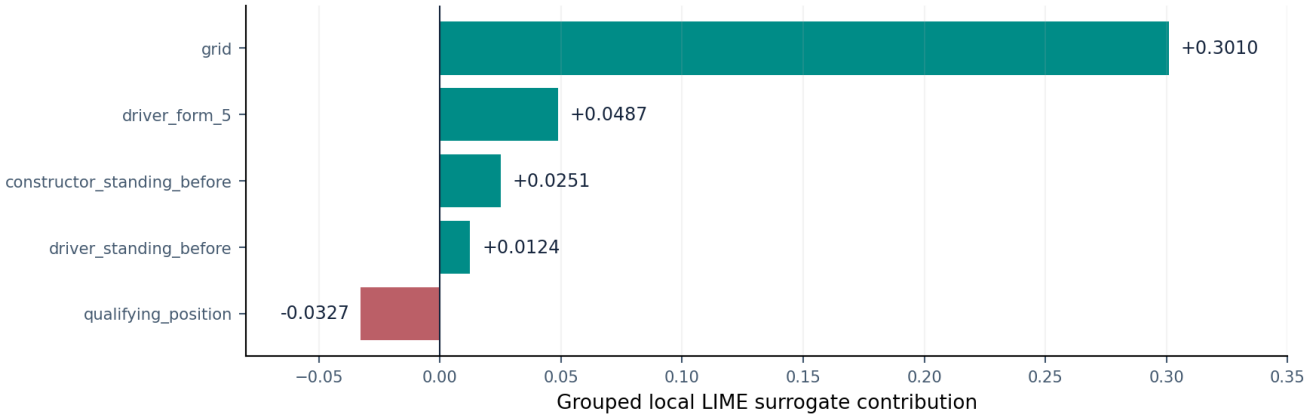


Figure 17. Local LIME contributions for the first successful execution, reconstructed from the original narrative.

LIME OBSERVATION	RECORDED RESULT
Predicted podium probability	75.23%
Observed podium outcome	1 (podium)
Largest positive contribution	Starting grid: +0.3010
Main negative contribution	Qualifying group: -0.0327

The selected validation driver-race has a predicted probability of 0.7523 in the original run and actually finished on the podium. Grid position strongly supports the prediction; qualifying position is locally negative. The LIME numbers are **local surrogate contributions**, not percentage-point causal effects.

8.4 Synthesis

EDA, feature-group ablation, permutation importance, SHAP and LIME all point toward the practical importance of starting grid. The methods answer different questions: permutation measures the impact of perturbing a trained model, SHAP distributes contributions around a background reference, LIME approximates one prediction, and ablation measures changes after retraining. Differences in secondary rankings are expected and should not be over-interpreted.

9. Reusable inference function

The notebook defines a predict_podium(record) function accepting the five original inputs, constructs a one-row DataFrame, applies the **previously fitted** 2B preprocessing pipeline, obtains the neural-network sigmoid probability and assigns a binary outcome using the network threshold learned on the validation set. No refitting occurs during inference.

9.1 Demonstration: complete versus partially missing

For the demonstration, the first complete observation from X_val.dropna() was used. The second scenario duplicates that record but hides its qualifying_position field. Both predictions are generated without error, demonstrating practical missing-value support.

INPUT FEATURE	COMPLETE	PARTIALLY MISSING
grid	3	3
qualifying_position	3	Missing
driver_form_5	0.8	0.8
driver_standing_before	2	2
constructor_standing_before	1	1

OUTPUT (FIRST EXECUTION)	COMPLETE	PARTIALLY MISSING
Podium probability	59.76%	53.91%
Predicted class	1 (podium)	1 (podium)

With qualifying hidden, the original-run probability decreased by **5.85 percentage points**, but both cases remained classified as podium outcomes using the validation-selected threshold. The complete and missing cases demonstrate a missingness-handling pipeline, not a claim that hiding qualifying information necessarily decreases probabilities for all drivers.

10. Limitations and reproducibility

10.1 Coverage, domain shift and feature interpretation

- **Historical availability:** qualifying data are missing in 16,174 of 26,668 driver-race rows (60.65%), and missingness changes across eras.
- **Era sensitivity:** racing regulations, number of starters, circuits and team identities vary widely from 1950 to 2024; pre-2019 patterns may not generalize uniformly.
- **Feature redundancy:** starting grid and qualifying position correlate strongly; coefficient signs and importance rankings can be unstable under correlated inputs.
- **Measurement:** free-text mechanical retirement status is a heuristic classification; nationality-to-country mapping omits unmatched labels.
- **Model selection:** hyperparameters were practical baselines; no confidence intervals, multiple-seed aggregates or external-season replication were established.
- **Task scope:** this is podium-versus-non-podium classification, not an exact finishing-order model or a causal analysis.

10.2 First execution versus subsequent Run All

The accompanying notebook explicitly states that its fixed numerical interpretations refer to the **first successful execution** and were retained unchanged after rerunning every cell. This report follows the same convention: numeric result tables and reconstructed experiment charts reflect that documented first run. Unchanged underlying data/cleaning outputs coincide across runs; stochastic model outputs may differ.

> **Important rerun-specific observation**
>
> The saved later execution records **neural network validation PR-AUC 0.7361**, slightly below **Random Forest 0.7372**. Thus, if model selection is recomputed strictly from the later Run All output, Random Forest becomes the best model on the primary validation metric. The neural-network-centered XAI and this report document the original selected network and must not be presented as selection from the later scores.

The rerun also reported a 72.84% LIME example probability and complete/missing inference probabilities of 61.61%/55.43%, while the first-run documented probabilities were 75.23% and 59.76%/53.91%. Later-run SHAP and training-curve images appear only as visual illustrations and are captioned accordingly. Reporting both avoids concealing the discrepancy while honoring the original notebook narratives.

11. Conclusion

A leakage-conscious driver-race dataset was engineered from 14 Formula 1 tables; all three relational questions were answered with joined data and visual evidence. Five original pre-race features, a chronological holdout protocol and missing-indicator preprocessing support Logistic Regression, Random Forest and shallow-neural-network comparisons. In the first successful run, the neural network narrowly led validation PR-AUC (0.7393) and reached 0.7131 on unseen test seasons. Across exploratory and explainability methods, starting grid emerged as the most influential input. The principal limitations are historical coverage drift, era changes and stochastic training variability.

References and artifact provenance

[1] Rohan Rao, *Formula 1 World Championship (1950–2020)*, Kaggle / Ergast-derived CSV dataset, used here with records through 2024. https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

[2] ACL course, *Milestone 1: Formula 1 Podium Prediction Companion*, task and deliverable requirements (provided in the project).

[3] Submitted Kaggle notebook, *acl-ms1-f1-podium(2).ipynb*, 81 cells; source for code, original-execution narratives, saved run results and notebook figures.

Figure provenance: Figures 1–9, 11–12 and 16 were extracted from the saved notebook outputs. Figures 10, 13–15 and 17 were recreated solely from the documented first-run numerical values to keep those charts consistent with the reported first-run interpretations.